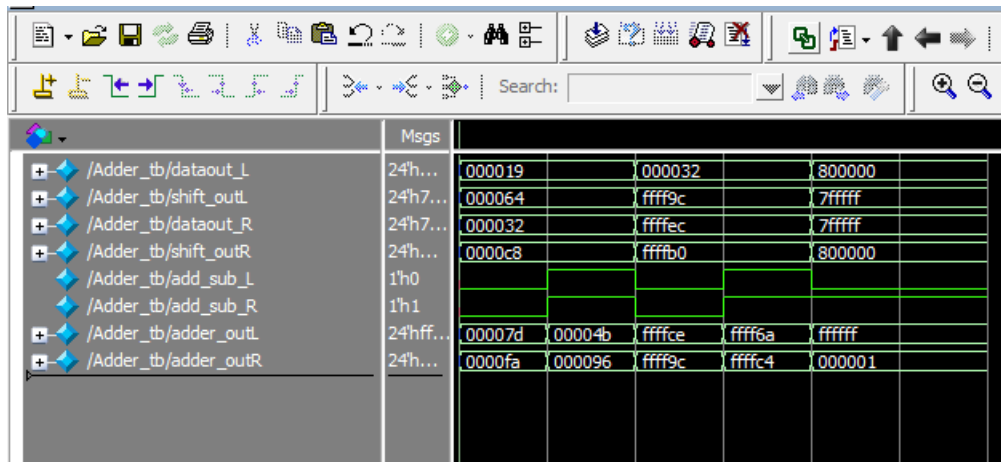


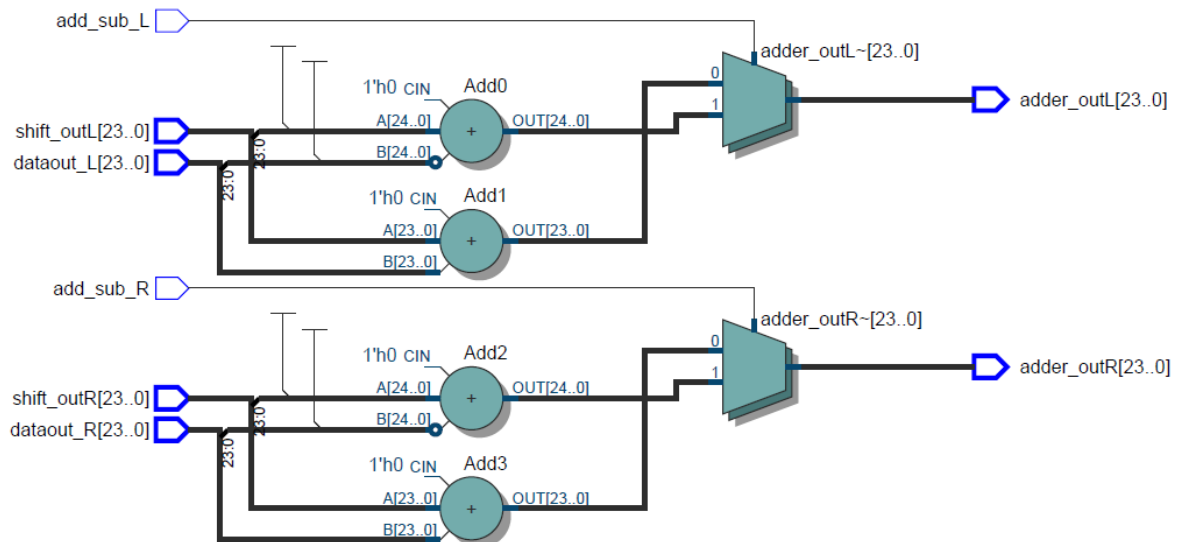
Adder block

This is two block adder meaning it add/subtract on both the right and Left side independently. The Block will take input from the SignExtender and from the 1-bitShiftRegister. It will use the MSB of the Coefficient at that stage of the filter to determine whether to perform addition or subtraction. The Block also has two independent outputs adder_outL/R

WaveView: showing sample for both addition and subtraction



Block Diagram of Adder: shows there are two blocks inside the Adder one for the right computation and one for the Left computation



Testbench table: showing values in decimal for easy recognition

```
# vsim -t lps -L altera_ver -L lpm_ver -L sgate_ver -L altera_mf_ver -L altera_insim_ver -L cyclonev_ver -L cyclonev_hssi_ver -L cyclonev_pcie_hip_ver -L rtl_work -L work -voptargs="+acc" Adder_tb
# vsim -t lps -L altera_ver -L lpm_ver -L sgate_ver -L altera_mf_ver -L altera_insim_ver -L cyclonev_ver -L cyclonev_hssi_ver -L cyclonev_pcie_hip_ver -L rtl_work -L work -voptargs="+acc" Adder_tb
# Start time: 23:55:15 on Mar 26, 2025
# ** Note: (vsim-3012) Design is being optimized...
# Loading work.Adder_tb(fast)
# Loading work.Adder(fast)
#
# add wave *
# view structure
# .main_pane.structure.interior.cs.body.struct
# view signals
# .main_pane.objects.interior.cs.body.tree
# run -all
Starting Adder Testbench...
Time=0 | L: 100 + 25 = 125 | R: 200 + 50 = 250
Time=10000 | L: 100 - 25 = 75 | R: 200 - 50 = 150
Time=20000 | L: -100 + 50 = -50 | R: -80 + -20 = -100
Time=30000 | L: -100 - 50 = -150 | R: -80 - -20 = -60
Time=40000 | L: 8388607 + -8388608 = -1 | R: -8388608 - 8388607 = 1
Testbench finished.
# ** Note: $finish : C:/Users/lmnmj/OneDrive/Documents/Grad School/UID/Spring 2025/CE 6306 ASIC Design/Projects/Midterm/ALU_Block/Adder/Adder_tb.v(74)
# Time: 60 ns Iteration: 0 Instance: /Adder_tb
# 1
```